الأسئلة المقالية — 9 عام · مذاكرة الهيكل الاختبار النهائي للفصل الاول

الاسم: _____ المدرسة: _____ التاريخ: _____ (65 سؤالاً)

LO21 إيجاد قيم التعابير الجبرية · Evaluate algebraic expressions

21.1 LO21 · Reveal p. 19 · 42

Evaluate if g=2, r=3, and t=11.

$$g + 6t$$

a. Show substitutions and evaluate. (10)

الأستاذ كامل بشارة 0502837825

21.2 LO21 · Reveal p. 19 · 43

Evaluate if g=2, r=3, and t=11.

$$7 - gr$$

a. Show substitutions and evaluate. (10)

الأستاذ كامل بشارة 0502837825

21.3 LO21 · Reveal p. 19 · 44

Evaluate if g=2, r=3, and t=11.

$$r^2 + (g^3 - 8)^5$$

a. Show substitutions and evaluate. (10)

الأستاذ كامل بشارة 0502837825

21.4 LO21 · Reveal p. 19 · 45

Evaluate if g=2, r=3, and t=11.

$$(2t + 3g) \div 4$$

a. Show substitutions and evaluate. (10)

21.5 LO21 · Reveal p. 19 · 46

Evaluate the expression if g=2, r=3, t=11. Show your substitutions and order of operations.

$$t^2 + 8rt + r^2$$

a. Show all steps and check your answer.

21.6 LO21 · Reveal p. 19 · 47

Evaluate the expression if g=2, r=3, t=11. Show your substitutions and order of operations.

$$3g(g+r)^2 - 1$$

a. Show all steps and check your answer.

21.7 LO21 · Reveal p. 19 · 48

Evaluate the expression if a=8, b=4, c=16. Show your substitutions and order of operations.

$$a^2bc - b^2$$

a. Show all steps and check your answer.

21.8 LO21 · Reveal p. 19 · 49

Evaluate the expression if a=8, b=4, c=16. Show your substitutions and order of operations.

$$\frac{a^2}{b^2} + \frac{b^2}{a^2}$$

a. Show all steps and check your answer.

21.9 LO21 · Reveal p. 19 · 50

Evaluate the expression if a=8, b=4, c=16. Show your substitutions and order of operations.

$$\frac{2b+3c^2}{4a^2-2b}$$

a. Show all steps and check your answer.

22.1 LO22 · Reveal p. 125 · 19

The perimeter of a rectangle is P=2ℓ+2w.

$$P = 2\ell + 2w$$

a. Solve for ℓ. (5)

b. Find ℓ when w=4 m and P=36 m. (5)

22.2 LO22 · Reveal p. 125 · 20

A batting average is a=h/b, with h hits in b times at bat.

$$a = h/b$$

a. Solve for b. (5)

b. Find b when a=0.325 and h=39. (5)

22.3 LO22 · Reveal p. 125 · 21

Videogames cost $13.50 each. Controllers cost p dollars; c is the total cost.

$$c = 13.50g + p$$

a. Solve for g. (5)

b. Find g when p=$55 and c=$136. (5)

22.4 LO22 · Reveal p. 125 · 22

The volume of a box is V=ℓwh.

$$V = \ell wh$$

a. Solve for h. (5)

b. Find h when V=50 m³, ℓ=10 m, and w=2 m. (5)

23.1 LO23 · Reveal p. 97 · 7

Solve the equation. Check your solution.

$$2(r+6)=4(r+4)$$

a. Show your steps. State if there is one solution, no solution, or an identity. (10)

23.2 LO23 · Reveal p. 97 · 11

Solve the equation. Check your solution.

$$3(3m-2)=2(3m+3)$$

a. Show your steps. State if there is one solution, no solution, or an identity. (10)

23.3 LO23 · Reveal p. 99 · 37

Solve the equation. Check your solution.

$$2x=2(x-3)$$

a. Show your steps. State if there is one solution, no solution, or an identity. (10)

23.4 LO23 · Reveal p. 99 · 39

Solve the equation. Check your solution.

$$-5(3-q)+4=5q-11$$

a. Show your steps. State if there is one solution, no solution, or an identity. (10)

23.5 LO23 · Reveal p. 96 · Example 6

Solve the equation. Check your solution.

$$6\,(y-5)=2\,(3y+10)$$

a. Show all steps and check your answer.

23.6 LO23 · Reveal p. 96 · Example 7

Solve the equation. Check your solution.

$$7x + 5\left(x - 1\right) = 12x - 5$$

a. Show all steps and check your answer.

23.7 LO23 · Reveal p. 97 · 8

Solve the equation. Check your solution.

$$6\left(n + 5\right) = 3\left(n + 16\right)$$

a. Show all steps and check your answer.

23.8 LO23 · Reveal p. 97 · 9

Solve the equation. Check your solution.

$$5\left(g+8\right)-7=117-g$$

a. Show all steps and check your answer.

23.9 LO23 · Reveal p. 97 · 10

Solve the equation. Check your solution.

$$12-\frac{4(x+15)}{5}=\frac{2x}{5}+6$$

a. Show all steps and check your answer.

23.10 LO23 · Reveal p. 97 · 12

Solve the equation. Check your solution.

$$6\left(3a+1\right)-30=3\left(2a-4\right)$$

a. Show all steps and check your answer.

23.11 LO23 · Reveal p. 97 · 13

Solve the equation. Check your solution.

$$7n+6=4n-9$$

a. Show all steps and check your answer.

23.12 LO23 · Reveal p. 97 · 14

Solve the equation. Check your solution.

$$-6\left(2r+8\right)=-10\left(r-3\right)$$

a. Show all steps and check your answer.

23.13 LO23 · Reveal p. 97 · 15

Solve the equation. Check your solution.

$$5-3\left(w+4\right)=w-7$$

a. Show all steps and check your answer.

23.14 LO23 · Reveal p. 97 · 16

Solve the equation. Check your solution.

$$2x - 5\left(x - 3\right) = 2\left(x - 10\right)$$

a. Show all steps and check your answer.

23.15 LO23 · Reveal p. 98 · 25

Solve the equation. Check your solution.

$$-6y - 3 = 3 - 6y$$

a. Show all steps and check your answer.

23.16 LO23 · Reveal p. 98 · 26

Solve the equation. Check your solution.

$$\frac{x+6}{2} = \frac{x}{2} - 9$$

a. Show all steps and check your answer.

23.17 LO23 · Reveal p. 98 · 27

Solve the equation. Check your solution.

$$8q + 12 = 4\,(2q + 3)$$

a. Show all steps and check your answer.

23.18 LO23 · Reveal p. 98 · 28

Solve the equation. Check your solution.

$$-6x + 21\,(x + 1) = 15x + 21$$

a. Show all steps and check your answer.

23.19 LO23 · Reveal p. 98 · 29

Solve the equation. Check your solution.

$$-4y + 12y + 48 = 8\,(y - 6)$$

a. Show all steps and check your answer.

23.20 LO23 · Reveal p. 98 · 30

Solve the equation. Check your solution.

$$8\left(z+6\right)=4\left(2z+12\right)$$

a. Show all steps and check your answer.

23.21 LO23 · Reveal p. 98 · 31

Solve the equation. Check your solution.

$$2a+2=3\left(a+2\right)$$

a. Show all steps and check your answer.

23.22 LO23 · Reveal p. 98 · 32

Solve the equation. Check your solution.

$$\frac{x}{4} + 5 = \frac{x}{4}$$

a. Show all steps and check your answer.

23.23 LO23 · Reveal p. 98 · 33

Solve the equation. Check your solution.

$$7\left(c + 9\right) = 7c + 63$$

a. Show all steps and check your answer.

23.24 LO23 · Reveal p. 98 · 34

Solve the equation. Check your solution.

$$4k + 3 = \frac{8k+16}{4}$$

a. Show all steps and check your answer.

23.25 LO23 · Reveal p. 98 · 35

Solve the equation. Check your solution.

$$3b + 4b - 13 = 7b + 1$$

a. Show all steps and check your answer.

23.26 LO23 · Reveal p. 98 · 36

Solve the equation. Check your solution.

$$\frac{\frac{m}{2}-8}{2} = \frac{m-16}{4}$$

a. Show all steps and check your answer.

23.27 LO23 · Reveal p. 99 · 38

Solve the equation. Check your solution.

$$\frac{2h}{5} - 7 = -2h + \frac{12h}{5} + 3$$

a. Show all steps and check your answer.

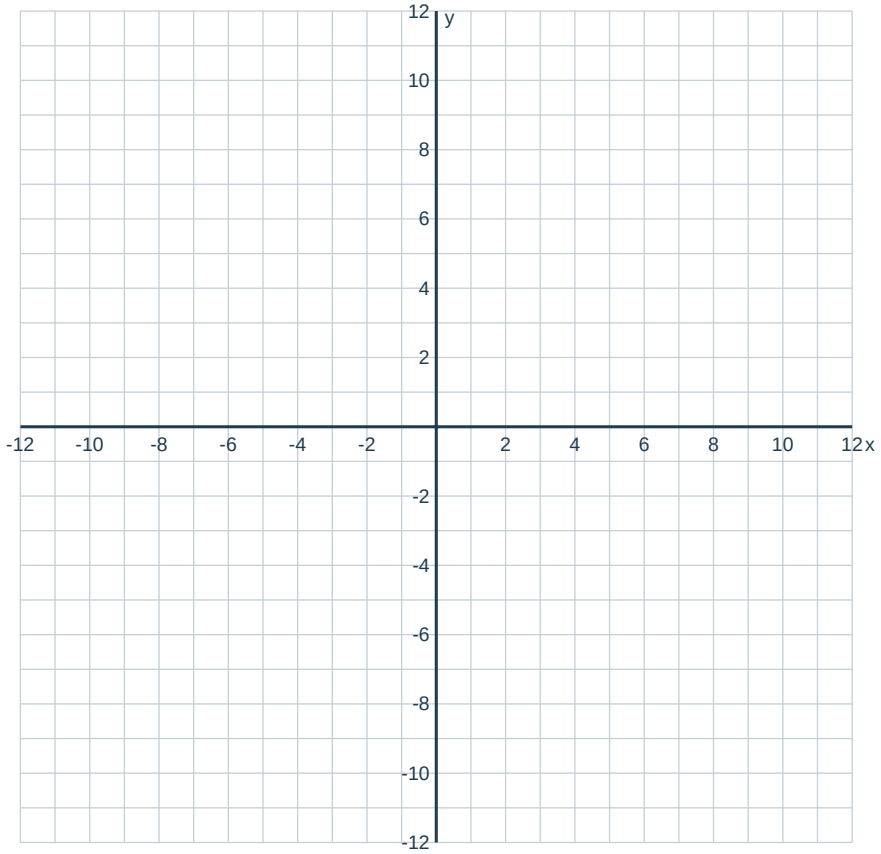
24.1 LO24 · Reveal p. 215 · 5

Graph the equation by making a table.

$$y - 8 = -x$$

a. Make a table of values, then plot the points and draw the complete line. (10)

x			
y			



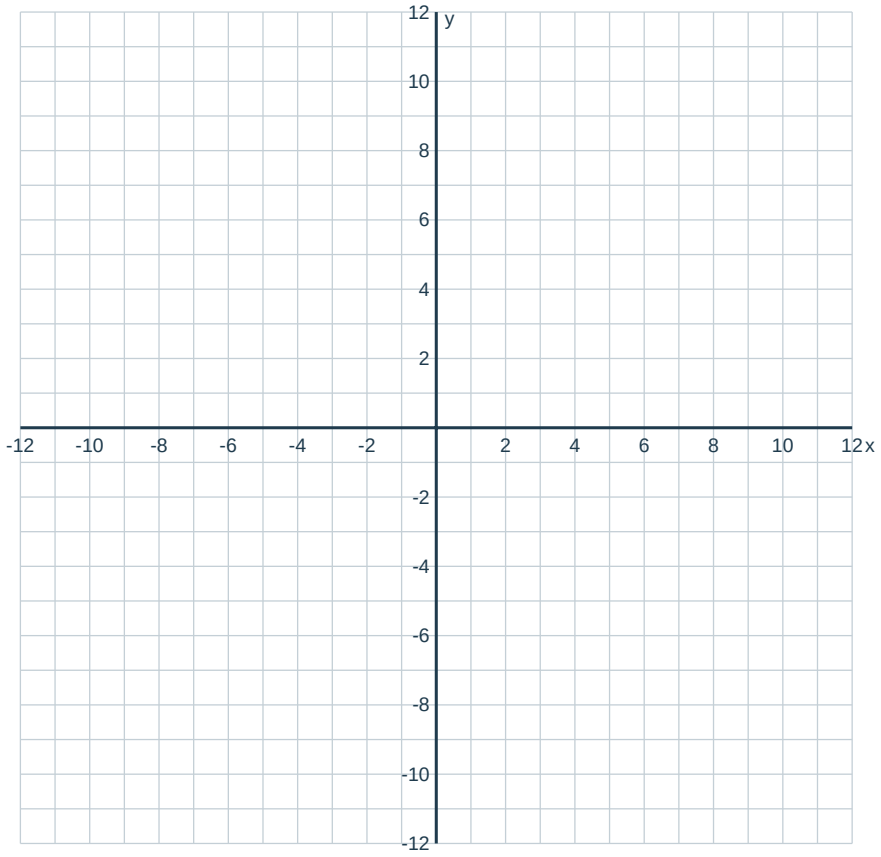
24.2 LO24 · Reveal p. 215 · 6

Graph the equation by making a table.

$$x = 10 - y$$

a. Make a table of values, then plot the points and draw the complete line. (10)

x			
y			



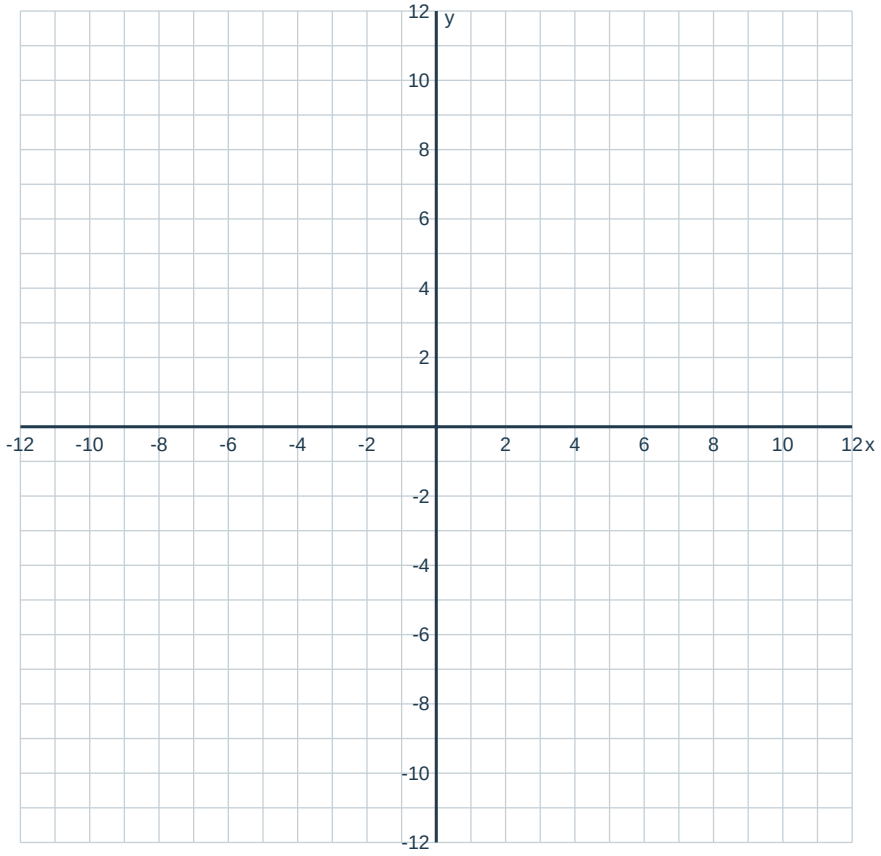
24.3 LO24 · Reveal p. 215 · 7

Graph the equation by making a table.

$$y = \frac{1}{2}x + 1$$

a. Make a table of values, then plot the points and draw the complete line. (10)

x			
y			



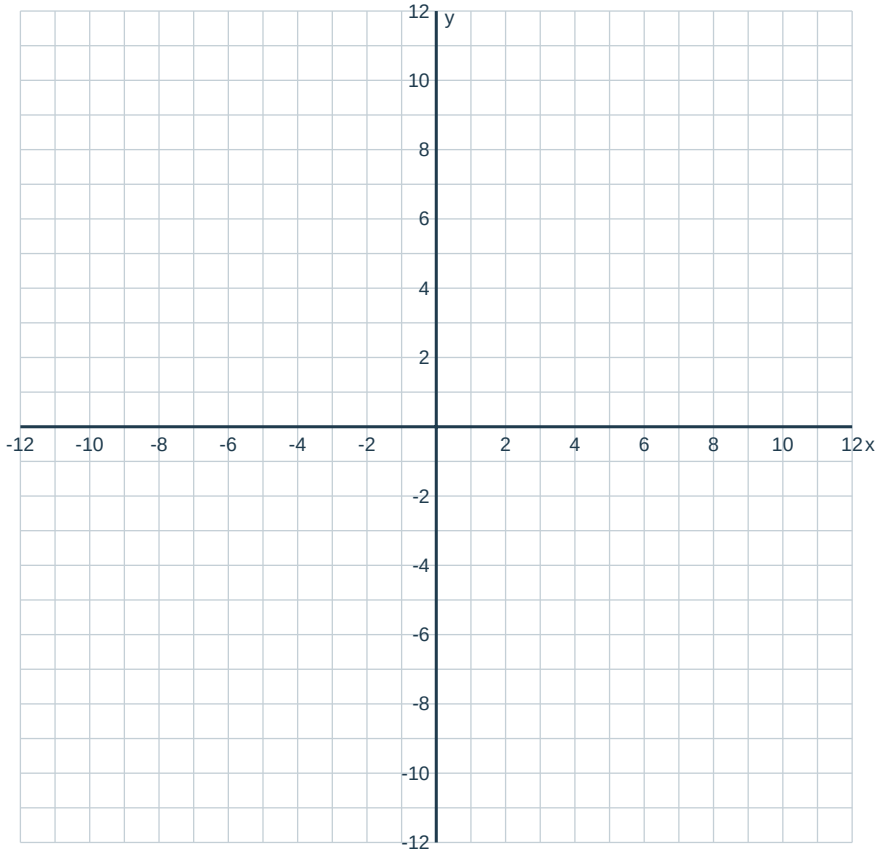
24.4 LO24 · Reveal p. 215 · 8

Graph the equation by making a table.

$$y + 2 = \frac{1}{4}x$$

a. Make a table of values, then plot the points and draw the complete line. (10)

x			
y			



24.5 LO24 · Reveal p. 215 · variant

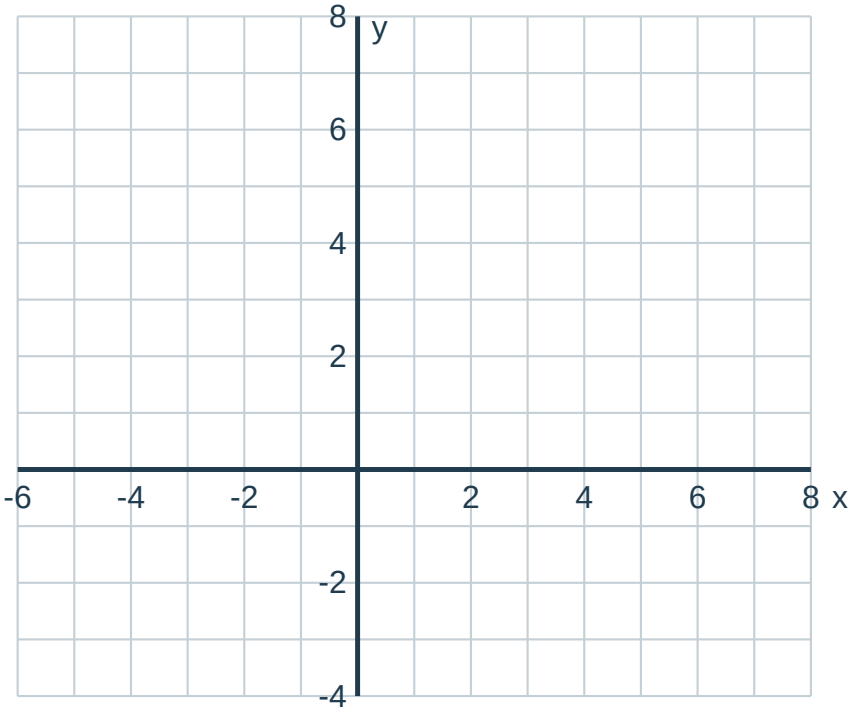
Graph the equation by making a table of values.

$$y = -\frac{1}{2}x + 3$$

a. Complete a table with three points.

b. Plot and draw the complete line.

x			
y			



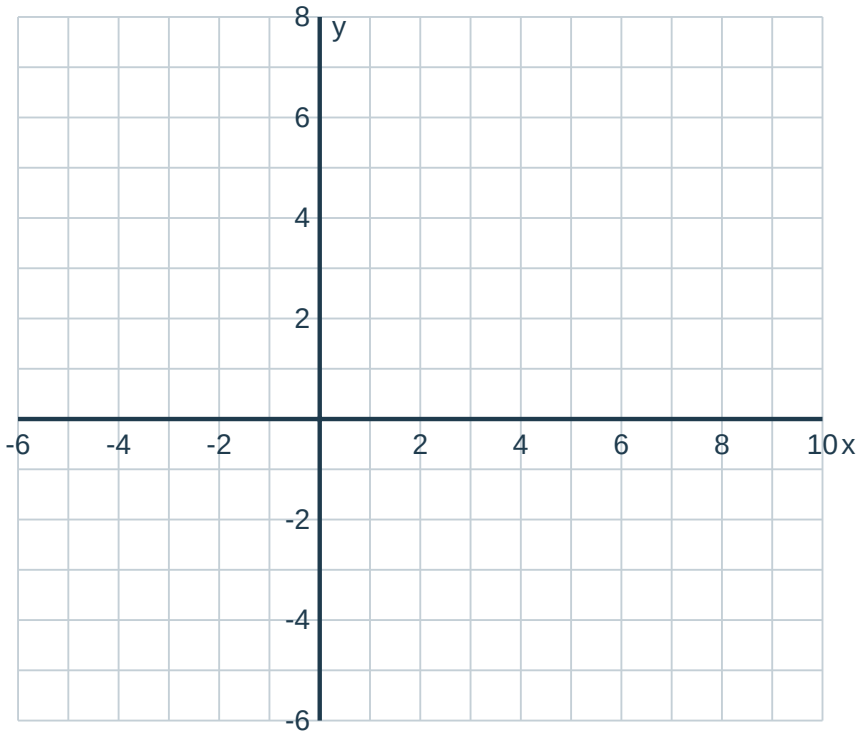
LO25 الدوال متعددة التعريف · Piecewise functions

25.1 LO25 · Reveal p. 263 · 1

Graph the function. State its domain and range.

$$f(x) = \begin{cases} \frac{1}{2}x - 1 & x > 3 \\ -2x + 3 & x \leq 3 \end{cases}$$

a. Plot both pieces with correct endpoints; state domain and range. (10)

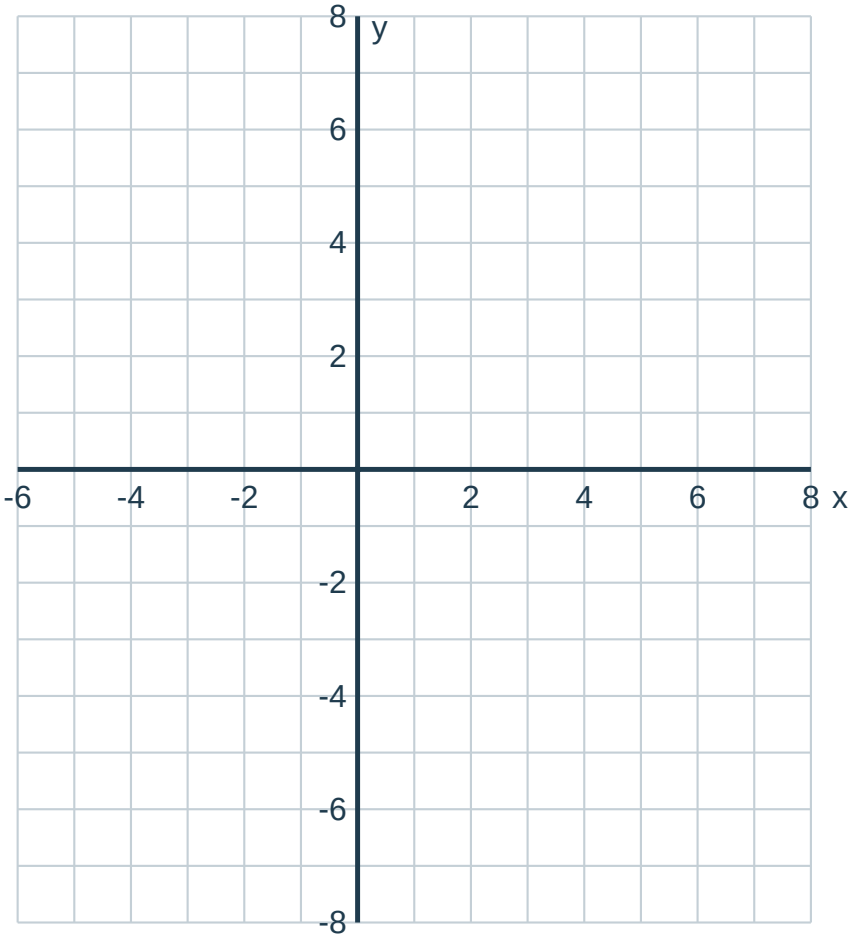


25.2 LO25 · Reveal p. 263 · 2

Graph the function. State its domain and range.

$$f(x) = \begin{cases} 2x - 5 & x > 1 \\ 4x - 3 & x \leq 1 \end{cases}$$

a. Plot both pieces with correct endpoints; state domain and range. (10)

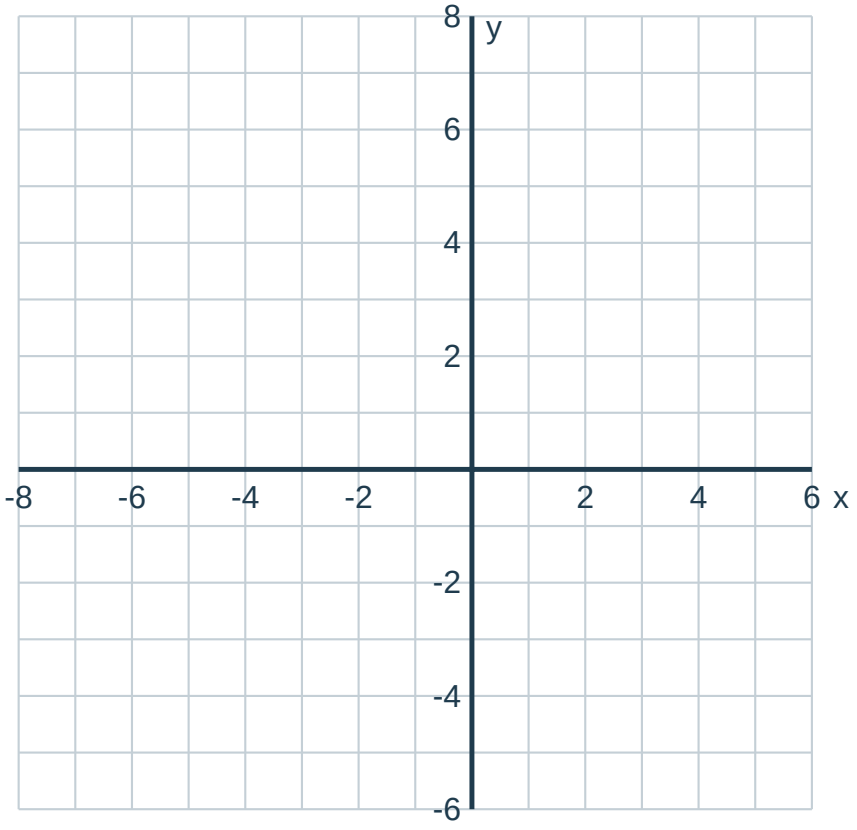


25.3 LO25 · Reveal p. 263 · 3

Graph the function. State its domain and range.

$$f(x) = \begin{cases} 2x + 3 & x \geq -3 \\ \frac{1}{3}x + 1 & x < -3 \end{cases}$$

a. Plot both pieces with correct endpoints; state domain and range. (10)

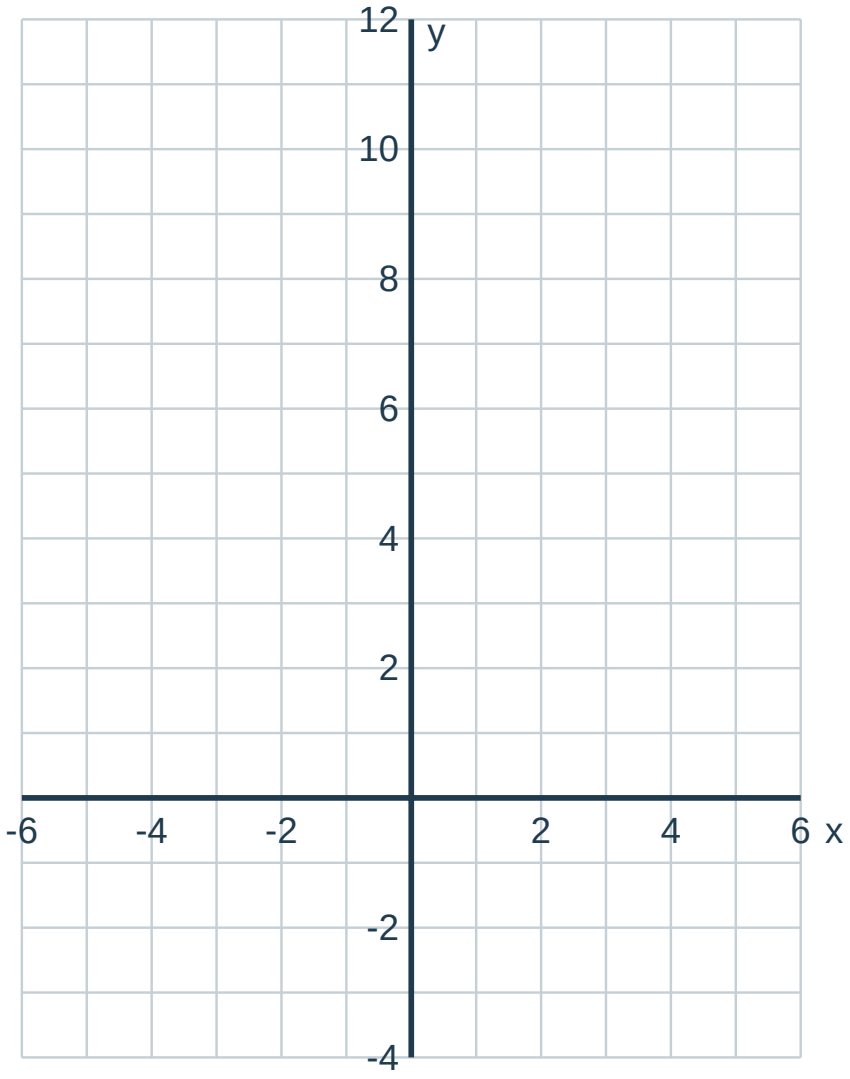


25.4 LO25 · Reveal p. 263 · 4

Graph the function. State its domain and range.

$$f(x) = \begin{cases} 3x + 4 & x \geq 1 \\ x + 3 & x < 1 \end{cases}$$

a. Plot both pieces with correct endpoints; state domain and range. (10)

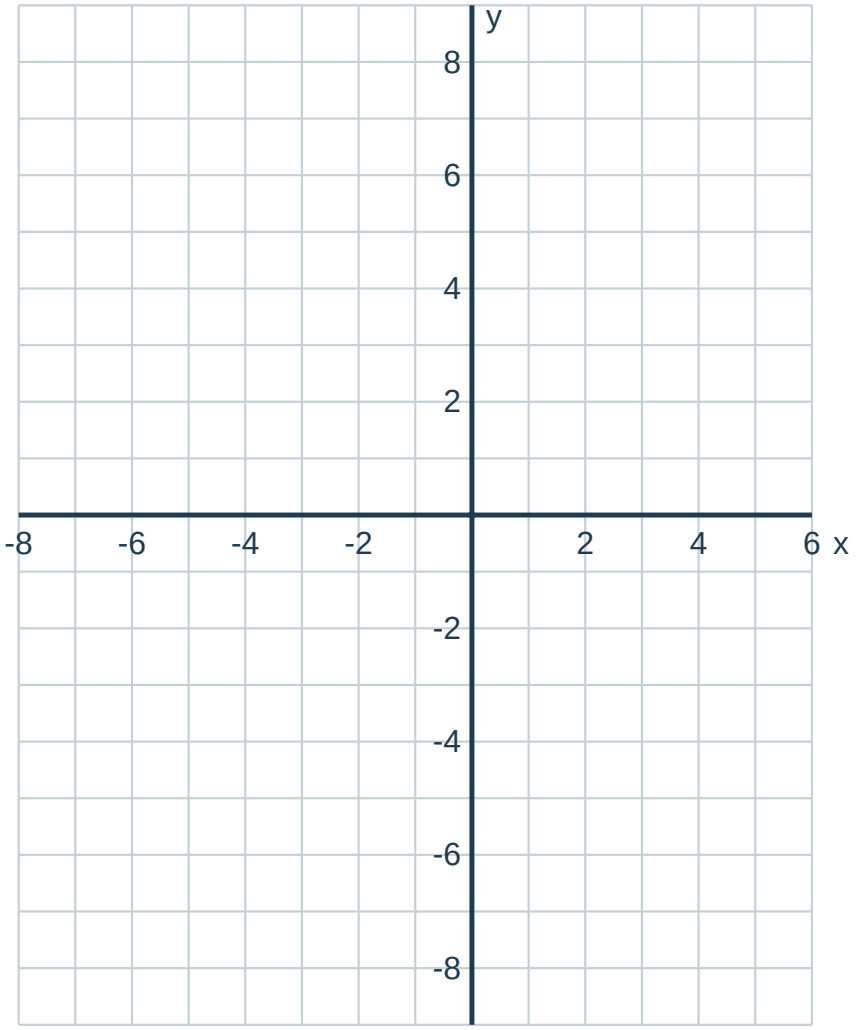


25.5 LO25 · Reveal p. 263 · 5

Graph the piecewise-defined function. State its domain and range.

$$f(x) = \begin{cases} 3x + 2 & x > -1 \\ -\frac{1}{2}x - 3 & x \leq -1 \end{cases}$$

a. Graph both pieces with the correct endpoints.



b. State the domain.

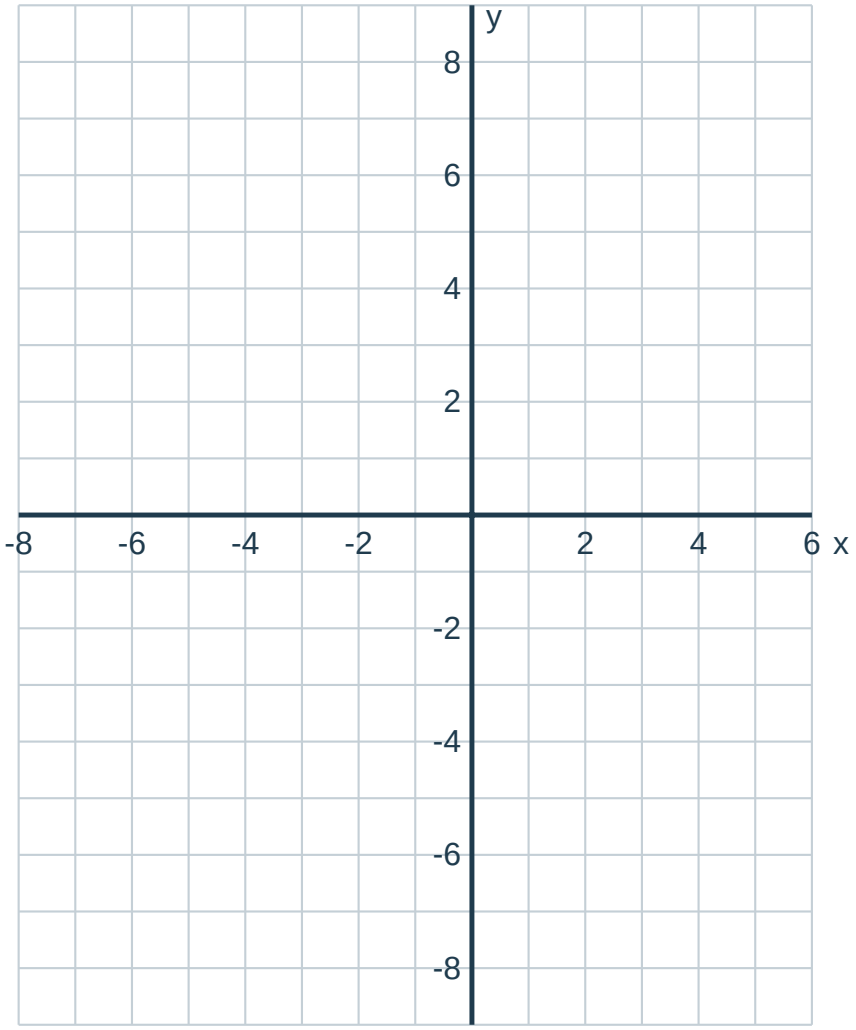
c. State the range.

25.6 LO25 · Reveal p. 263 · 6

Graph the piecewise-defined function. State its domain and range.

$$f(x) = \begin{cases} 2x + 1 & x < -2 \\ -3x - 1 & x \geq -2 \end{cases}$$

a. Graph both pieces with the correct endpoints.



b. State the domain.

c. State the range.

26.1 LO26 · Reveal p. 255 · 19

Use the arithmetic sequence to write an equation and find the 7th term.

$$-3, -8, -13, -18, \ldots$$

a. Write a formula for the nth term. (5)

b. Find the 7th term. (5)

26.2 LO26 · Reveal p. 255 · 20

Use the arithmetic sequence to write an equation and find the 7th term.

$$-2, 3, 8, 13, \ldots$$

a. Write a formula for the nth term. (5)

b. Find the 7th term. (5)

26.3 LO26 · Reveal p. 255 · 21

Use the arithmetic sequence to write an equation and find the 7th term.

$$-11, -15, -19, -23, \ldots$$

a. Write a formula for the nth term. (5)

b. Find the 7th term. (5)

26.4 LO26 · Reveal p. 255 · 22

Use the arithmetic sequence to write an equation and find the 7th term.

$$-0.75, -0.5, -0.25, 0, \ldots$$

a. Write a formula for the nth term. (5)

b. Find the 7th term. (5)

26.5 LO26 · Reveal p. 255 · 9

Find the common difference of the arithmetic sequence, then find the next three terms.

$\frac{1}{50}, \frac{27}{25}, \frac{107}{50}, \frac{16}{5}, \ldots$

a. Find the common difference.

b. Find the next three terms.

26.6 LO26 · Reveal p. 255 · 10

Find the common difference of the arithmetic sequence, then find the next three terms.

$6,\ 12,\ 18,\ 24, \ldots$

a. Find the common difference.

b. Find the next three terms.

26.7 LO26 · Reveal p. 255 · 11

Find the common difference of the arithmetic sequence, then find the next three terms.

$21,\ 19,\ 17,\ 15, \ldots$

a. Find the common difference.

b. Find the next three terms.

26.8 LO26 · Reveal p. 255 · 12

Find the common difference of the arithmetic sequence, then find the next three terms.

$-\frac{1}{2},\ 0,\ \frac{1}{2},\ 1, \ldots$

a. Find the common difference.

b. Find the next three terms.

26.9 LO26 · Reveal p. 255 · 13

Find the common difference of the arithmetic sequence, then find the next three terms.

$\frac{7}{3}, \frac{8}{3}, 3, \frac{10}{3}, \ldots$

a. Find the common difference.

b. Find the next three terms.

26.10 LO26 · Reveal p. 255 · 14

Find the common difference of the arithmetic sequence, then find the next three terms.

$\frac{7}{12}, \frac{4}{3}, \frac{25}{12}, \frac{17}{6}, \ldots$

a. Find the common difference.

b. Find the next three terms.

26.11 LO26 · Reveal p. 255 · 15

Find the common difference of the arithmetic sequence, then find the next three terms.

$3,\ 7,\ 11,\ 15, \ldots$

a. Find the common difference.

b. Find the next three terms.

26.12 LO26 · Reveal p. 255 · 16

Find the common difference of the arithmetic sequence, then find the next three terms.

$22,\ \frac{39}{2},\ 17,\ \frac{29}{2}, \ldots$

a. Find the common difference.

b. Find the next three terms.

26.13 LO26 · Reveal p. 255 · 17

Find the common difference of the arithmetic sequence, then find the next three terms.

$-13,\ -11,\ -9,\ -7, \ldots$

a. Find the common difference.

b. Find the next three terms.

26.14 LO26 · Reveal p. 255 · 18

Find the common difference of the arithmetic sequence, then find the next three terms.

$-2,\ -5,\ -8,\ -11, \ldots$

a. Find the common difference.

b. Find the next three terms.